The Effects of Refugees' Emotional Tears on Felt Emotions and Helping Behaviors

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Data availability

Data, code, and materials from this research are available at https://osf.io/86rqg/?view_only=f5cfc624cfbd4ff88766a1f5ce7ac068 (anonymous link with access to all components of the project).

Note. Preprint ahead of peer-review.

Scientific articles usually go through a peer-review process. This means that independent researchers evaluate the quality of the work, provide suggestions and speak for or against the publication. Please note that the present article has not (yet) undergone this standard procedure for scientific publications.

Abstract

Tearful faces are frequent in mass media portraits of refugees, but the consequences of such depictions for different kinds of intergroup helping remain poorly understood. Moreover, existing research suggests that individuals tend to be more *willing* to support those who display tears, yet the impact of emotional tears on *actual* helping behavior has not been examined. Across three experiments relying on nation-wide panel data ($N = 2647$) and a quasi-experimental citizen science study ($N = 191$) conducted in the Netherlands, we tested the effects of refugees' tearful (vs. non-tearful) expressions on people's emotional reactions, intentions to offer refugees both dependency- and autonomy-oriented help, as well as actual helping behavior (i.e., money donated to an organization supporting refugees). Data from three online experiments unveiled a positive relationship between exposure to emotional tears and both dependency- and autonomy-oriented intentions to help, as well as donation behavior (in Study 4, two initiatives encompassing both dependency- and autonomy-oriented help), via felt compassion. Across all studies, the visual representations of refugees with tears consistently resulted in more helping intentions and donations indirectly via the perceived sadness of the refugee(s) and the subsequent felt compassion. Our findings illuminate critical questions regarding visual representations of refugees that might motivate solidarity. They also encourage future collaborative research at the convergence of art, social and emotion science, and activism. Our research also provides a platform for initiating fundraising campaigns aimed at generating resources to support initiatives focused on sharing the narratives of displaced individuals and empowering the refugee community.

Key words: refugees, emotional tears, compassion, anger, intergroup helping, donation

The Effects of Refugees' Emotional Tears on Felt Emotions and Helping Behaviors

Being forced to flee one's homeland involves severe perils and suffering. The loss of close ones, separation from one's family, physical assault, and loss of property are among the most painful experiences refugees are faced with (e.g., Echterhoff et al., 2020). In such dire circumstances, people typically feel helpless and frequently cry (Vingerhoets, 2013; Vingerhoets & Bylsma, 2016). It is not surprising that tearful images are common portrayals of forced migrants in mass and social media.[1] Yet, the impact of these tearful refugee portrayals on the receiving society remains poorly understood. People are generally less motivated to care about and help outgroup members compared to ingroup members (Cikara et al., 2011), which makes it important to understand what drives solidarity with refugees. Their facial emotion expressions, such as crying, might constitute one relevant factor.

With three online experiments employing nation-wide samples and one quasi-experimental field study integrated into a real-world photography exhibition, our research systematically examined the influence of tearful images of refugees on intergroup helping. Our contribution to the literature on emotions and intergroup relations encompasses several facets: a) investigating the inter*group* prosocial implications of tearful displays, b) focusing on two different forms of intergroup helping intentions (i.e., dependency- and autonomy-oriented help), c) examining the impact of tears on tangible intergroup helping behavior (i.e., donations to initiatives supporting refugees), and d) delving into the role of emotional reactions (compassion, sadness, distress, and moral anger) within the connection between tears and intergroup helping.

Emotional Tears and Intergroup Helping

[1] For the purpose of this paper we use the terms "refugees", "forced migrants", and "asylum seekers" interchangeably.

Researchers argue that we cry emotional tears[2] when we feel incapable of dealing with a situation and feel overwhelmed and helpless (Vingerhoets, 2013). Crying typically accompanies emotions such as distress, sadness, anger, fear, or disappointment (Vingerhoets, 2013; Vingerhoets & Bylsma, 2016), and tears serve as signals for observers to gauge the crying person's helplessness and the intensity of their emotional state (Gračanin et al., 2021).[3] Empirical evidence strongly supports the notion that tears instill intentions to approach and support a crying person, across different cultures and situations (e.g., Gračanin et al., 2018; Hendriks et al., 2008; Hendriks & Vingerhoets, 2006; Zickfeld et al., 2021; see Zickfeld & Wróbel, 2023 for a review). Recent findings even indicate that this holds true for tearful expressions in immigrant individuals (Bobowik et al., 2022). So far, however, little attention has been paid to the role of tearful displays in stimulating *intergroup* prosociality, including charity giving and cooperation (e.g., with refugees as a group of people), rather than *interpersonal* social support such as offering comfort (e.g., to a specific refugee as a person in need).

An emerging body of literature recognizes the need to distinguish between various ways people express their care and concern for others (Louis et al., 2019; Nadler & Halabi, 2006; Thomas & McGarty, 2018), including refugees (Echterhoff et al., 2022). According to the Model of Intergroup Helping as Power Relations (e.g., Nadler & Halabi, 2006), help offered to refugees can take two main forms: *dependency-oriented help*, which provides immediate solutions to a problem (e.g., donating food or offering temporary shelter), and *autonomy-oriented help*, which offers resources enabling individuals to address problems by themselves (e.g., access to employment, housing or regular income). Previous research

[2] Researchers differentiate between three different types of tears: basal tears, tears that help to keep the eye lubricated, reflex tears, tears shed due to irritation to the eye, and emotional tears, tears shed in response to emotional reactions. Here, we focus on the latter.

[3] Apart from negative emotions, tears can also accompany positive emotions (see Zickfeld et al., 2020).

suggests that in response to tears, people are willing not only to provide immediate emotional support to immigrants but also to engage in charitable giving,such as donating to organizations that assist immigrants (Bobowik et al., 2022). However, this earlier research did not differentiate between dependency- and autonomy-oriented forms of helping. Hence, it remains unclear whether tearful representations encourage greater aid aimed at alleviating refugees' immediate suffering (i.e., more dependency-oriented help), and/or stronger support for refugees to become self-sufficient in improving their situation (i.e., autonomy-oriented help).

Additionally, existing research on the effects of emotional tears has primarily concentrated on behavioral intentions, with limited consideration of actual intergroup behavior (Gračanin et al., 2018; Reed et al., 2019 examined interpersonal behavior). Measuring meaningful real-life behavior presents a significant challenge in the social sciences, especially in the *intergroup* domain in which self-reports, laboratory-based responses, or resource-intensive observations are used (Baumeister et al., 2007). To the best of our knowledge, no prior study has examined the extent to which individuals genuinely engage in charitable acts towards refugees in response to their depictions with tears. Charitable giving represents a vital form of intergroup helping and a crucial aspect of prosocial behavior, alongside providing emotional comfort and fostering cooperation (Dovidio et al., 2017). Therefore, our research delved into the impact of refugees' tears on tangible actions: donating money to organizations that support refugees (e.g., by sharing their stories and thus amplifying their voices).

The Role of Felt Emotions

If emotional tears indeed enhance inter*group* helping, one important question is *why* this is the case. Research on the role of tears in an inter*personal* setting suggests that the tea-induced propensity for help involves affective processes such as empathic concern or

compassion (Bobowik et al., 2022; Zickfeld et al., 2021) and sadness (Küster, 2018). We contend that these discrete emotional reactions might be associated differently with intentions to engage in dependency-oriented or autonomy-oriented intergroup help. We thus examined whether various emotional reactions underlie the different intergroup helping tendencies that tears in depictions of refugees are expected to elicit. We focused on four different types of emotional reactions to tears: compassion and moral anger as other-oriented emotional responses, and distress and sadness as self-oriented emotional reactions.

Other-Oriented Felt Emotions: Compassion and Moral Anger

Compassion arises when individuals witness the undeserved suffering of others in need, prompting them to provide help to the suffering party (Goetz et al., 2010).[4] Research indicates that tears enhance the willingness to offer social support by eliciting compassionate emotional reactions (Zickfeld & Wróbel, 2023), including in response to tears from immigrants (Bobowik et al., 2022). The role of compassion in promoting helping may weaken when substantial costs are involved (Cialdini et al., 1997; Neuberg et al., 1997; Vitaglione & Barnett, 2003), but research in the field of intergroup relations suggests that feelings of compassion are linked also with outgroup prosociality, such as solidary collective action (e.g., Iyer et al. 2003; Saab et al., 2015).

Furthermore, a study involving Greek participants showed that compassionate feelings were linked with self-reported dependency-oriented helping of refugees (offering "food," "clothes," or "money") and, to a lesser extent, positive attitudes towards refugees' permanent settlement (Thravalou et al., 2021). Another study found that employees of charity- and human-rights organizations associate compassionate feelings with dependency-oriented aid towards refugees, such as contributing to projects that provide medical

[4] Various terms such as *empathic concern*, *empathy*, *sympathy*, *pity*, or *tenderness* have been used to refer to more or less the same emotional response as compassion (Batson et al., 1987; Zickfeld et al., 2017) or to slightly different phenomena (e.g., Lishner et al. 2011).

assistance, but not necessarily with autonomy-oriented helping intentions (Knab & Steffens, 2021). Further, Lantos et al. (2020, Study 2) observed that feelings of pity were more strongly associated with dependency-oriented helping tendencies (i.e., donating clothing, school supplies, toys, or money) than with autonomy-oriented ones (i.e., demonstrations or petitions demanding more rights) towards economically disadvantaged groups such as refugees. We examined whether tearful expressions by refugees activate compassionate feelings that particularly motivate dependency-oriented helping intentions, as well as actual donation behavior.

Additionally, Hendriks et al. (2008) observed that tearful displays often evoke strong negative emotions, including experiences of anger, which can coincide with distress. In Bobowik et al.'s study (2022), tearful (but also non-tearful) displays of sadness in immigrant faces activated reactions of anger, which might be associated with participants' annoyance towards the crying individuals (anger directed at the crier) or their anger regarding the suffering and injustice experienced by the refugees (anger directed at injustice). Moral or system-based anger typically arises when individuals witness a transgression towards others (i.e., a violation of equity), and thus is focused on the harm and its intentionality (e.g., Russell & Giner-Sorolla, 2011). Such emotions motivate a willingness to seek justice for the victim (e.g., Darley & Pittman, 2003; Haidt, 2003; Leach, 2008; van Doorn et al., 2014). There is evidence to suggest that moral anger motivates support for autonomy-oriented prosocial actions, including assisting victims in starting anew (e.g., van Doorn et al, 2017), offering them compensations in both fictitious (e.g., Van den Vyer & Abrams, 2015) and real-life scenarios (e.g., Iyer et al., 2007), or facilitating their access to higher education (e.g., Wakslak et al., 2007).

Several studies have shown that moral anger can be a relatively stronger predictor of autonomy-oriented helping compared to dependency-oriented helping. For instance, in the

study by Doorn et al. (2017), emotional reactions characterized by anger were more strongly associated with donations to a charity aimed at increasing autonomy (e.g., compensating the suffering of women so that they can start a new life) as compared to dependency-oriented financial contributions (e.g., alleviating women's suffering and not worsening their situation). Similarly, research shows that employees of charity- and human-rights entities often link feelings of moral outrage with their organization's goal of autonomy-promoting assistance to refugees, such as donating to projects designed to empower refugees to advocate for their rights, but not necessarily with dependency-oriented actions (Knab & Steffens, 2021). However, data from Germany and Hungary reveal that, when accounting for feelings of compassion (or pity), the relationship between moral anger and both types of help was either non-significant or weak (Lantos et al., 2020). Our research sought to investigate whether tearful expressions by refugees evoke feelings of moral anger, thereby motivating particularly autonomy-oriented helping intentions, as well as donation behavior.

Self-Oriented Felt Emotions: Distress and Sadness

In addition to compassion and moral anger, exposure to the suffering of others might elicit feelings of distress (Batson et al., 1983; 1997; Hendriks et al., 2008), and other largely unpleasant, self-oriented emotions such as sadness (Küster, 2018).[5] Some researchers have put forward that these self-oriented emotions might also drive people to offer help as a means to reduce their own negative feelings (e.g., Lench et al., 2016; Batson et al., 1997; Dovidio et al., 1997), although this tendency is mostly limited to relatively low-cost forms of assistance (e.g., Cialdini et al., 1997). Several studies have pointed towards the role of sadness (Küster, 2018) and distress (Zickfeld et al., 2021) as explanatory factors in the link between others' tears and willingness to offer immediate comfort, although other research has not consistently

[5] Research has provided evidence that distress and sadness are two distinct emotions in response to other's suffering most importantly differing in their arousal level (Fultz et al., 1988).

supported the role of felt distress (Bobowik et al., 2022). When considering felt compassion and distress as two emotional reactions simultaneously, both played a role but with compassion showing a much stronger indirect effect (Zickfeld et al., 2021). In our research, we expected that felt distress and sadness, along with compassion, can act as two explanatory mechanisms in the link between expressed tears and both dependency-oriented helping intentions and helping behavior.

The Present Research

Confirmatory Hypotheses

We experimentally tested the effects of tearful (vs. non-tearful) expressions of refugees on helping (i.e., autonomy- and dependency-oriented helping intentions and actual charity giving to a social initiative supporting refugees) and examined the roles of felt other-oriented emotions (compassion and moral anger) and self-oriented emotions (sadness and distress) as explanatory mechanisms.

We focused on testing the following main effects across all studies: We expected that tearful (vs non-tearful) expressions would increase feelings of compassion, sadness, distress, and moral anger (H1), lead to more dependency-oriented and autonomy-oriented helping intentions towards refugees (H2)[6], and to more donations to an initiative oriented at helping refugees (H3, in Study 4, we considered both types of initiatives: a dependency-oriented one and an autonomy-oriented one). In addition, we tested possible indirect processes via felt emotions. The effect of tearful displays on dependency-oriented helping intentions was expected to be explained by the increased feelings of compassion, sadness, and distress (H4), whereas its effect on autonomy-oriented helping intentions was expected to be explained by the increased moral anger (H5). The effects of tearful displays on donation behavior would be

[6] We also pre-registered a hypothesis concerning the strength of the effects of tears on autonomy-oriented helping intentions compared to dependency-oriented ones. However, the two types of help were quite strongly correlated and both effects were not statistically significant across the three studies.

mediated by compassion, sadness, distress, and moral anger (H6, in Study 4, considering both types of initiatives).

In Study 4, we additionally pre-registered two hypotheses with sequential effects. We expected that the exposure to tearful portrayals of refugees will lead to higher dependency-oriented helping behavior indirectly via higher felt compassion and consequently higher intentions to offer dependency-oriented help (H7), and to higher autonomy-oriented helping behavior indirectly via higher felt compassion and consequently higher intentions to offer autonomy-oriented help (H8).

Exploratory Research Questions

In addition to our preregistered hypotheses and in line with the notion that intentions predict behavior (Chandon et al., 2005; Fishbein & Ajzen, 1975), it is also possible that helping intentions towards refugees underly actual prosocial behaviors. We thus additionally explored a sequential mediation model in which the effects of tearful (vs non-tearful) expressions on donations are sequentially mediated by felt emotions (compassion, distress, sadness, and moral anger), which in turn predict dependency-oriented and autonomy-oriented helping intentions.[7]

Drawing on the empathy-altruism literature (Batson et al., 1981, 1987) and empirical evidence suggesting that sadness (e.g., Bobowik et al., 2022; van de Ven et al., 2017), and other emotions attributed to the crier (e.g., Zickfeld & Schubert, 2018), are explanatory mechanisms of the interpersonal effects of tears, we also estimated a sequential model in which the effects of tears on helping intentions and behavior occur indirectly via perceived sadness and felt compassion as two sequential mediators.

Considering the social-signaling function of multi-modal emotion expression

[7] We originally pre-registered these exploratory analyses as two separate models. However, for consistency with the main model, we tested a single model in which all emotions were introduced as parallel mediators, and dependency- and autonomy-oriented helping intentions as two other parallel mediators, predicted by the four emotions.

(Monroy et al., 2022; van Kleef, 2017), we further examined the effect of acoustic signals of crying in Study 1. Based on the available audiovisual material, we were able to create two different video clips with the same tearful expressions: a) a non-vocal condition in which the target is silent, and b) a vocal condition in which the target is speaking. We did not have specific predictions in relation to the addition of voice to the visual display of emotional tears. Still, it is possible that a facial expression with voice might have an accumulative effect on people's reactions to tears. In this case, the vocal tearful condition would have a stronger impact on helping refugees compared to the non-verbal tearful condition. We also tested if the effects of vocal vs. non-vocal tearful expressions on dependency- and autonomy-oriented helping intentions and donation behavior occurs indirectly via felt emotions.

Finally, we explored several possible moderation effects in the link between tears and outgroup helping, including interactions with gender, age, political ideology, and feelings of responsibility. This allowed us to test the robustness of the predicted relations across several individual difference variables.

We tested our predictions across four studies conducted in the Netherlands: three pre-registered experimental studies (Study 1: https://osf.io/jt75a, Study 2: https://osf.io/jsz76, Study 4: https://osf.io/kdjwh) relying on nation-wide, large samples of native-born participants, and one pre-registered field quasi-experiment involving a real-world photography exhibition (Study 3: https://osf.io/jhc3w). In the last decade, the accommodation of refugees has been put into the spotlight of the public and political debate in the Netherlands, resulting in a strong polarization in attitudes towards refugees, ranging from anti-refugee protests to volunteering in solidarity with them (e.g., Albada et al., 2021; Gootjes et al., 2021). All studies were approved by the ethics committee of the Dutch university in question. All participants gave their informed consent. Participation was voluntary, and participants were remunerated in Studies 1, 2, and 4. A complete version of the

questionnaires in Studies 1-4 in English and/or in Dutch can be found at the project page on OSF (https://osf.io/86rqg/). Experimental stimuli are available upon request.

Study 1

Study 1 was an online experiment in which Dutch participants were randomly exposed to one of three experimental conditions: a video of a refugee woman either not showing or showing emotional tears (with a vocal component and without). We tested our empirical model, in which felt emotions are the explanatory mechanism in the expected tears-helping link.[8]

Method

Participants

We set the smallest effect size of interest (SESOI, Lakens, 2022) to $d = 0.20$, based on existing empirical evidence (the average effects of tears on social support intentions in meta-analyses ranged from $d = 0.56$ to 0.69, Zickfeld et al., 2021, https://osf.io/48qjm/ Figures 1.2.1 and 4.6.1) and expecting a smaller effect when measuring behavior (e.g., Eisenberg & Miller, 1987). A power analysis using G*Power 3.1 (Faul et al., 2009) indicated that obtaining an effect size $d = 0.20$ when comparing two independent means, with an alpha of 0.05 and power of 0.8 would require 310 participants per experimental condition, so in total 930. Considering possible exclusions (e.g., due to a failed attention check), we stopped data collection when we surpassed the sample of 1,000 complete responses ($N = 1018$) after the attention check filter.[9]

Participants were recruited by the research company Motivaction that manages participant panels in the Netherlands. Panelists received a one-use-only online invitation to

[8] Before conducting Study 1, we implemented a pilot study in which the experimental manipulation showed a ceiling effect in perceived sadness. We therefore did not continue the data collection and revised the narrative presented with the video clips.

[9] Such sample size, with an alpha of 0.05 and power of 0.8, allows detecting effects as small as $\beta = .13$ in structural equation modeling with 6 latent variables and 22-15 observed variables (Soper, 2022).

participate in the research (to reduce self-selection risks and avoid duplicated responses). The recruitment company invited participants based on the following criteria: being minimum 18 years old, resident in the Netherlands for the last 10 years, born in the Netherlands, and both parents born in the Netherlands. A total of 1557 participants filled in the survey since we continued the data collection until reaching the expected sample size for complete responses. All participants who did not meet the conditions of the attention check included toward the end of the survey were screened out (n = 539 in total, including 386 who did not complete the survey and 153 participants who gave an incorrect response in the attention check item). After doing that, we checked if there were additional speeders based on Tier 3 criteria, that is, faster than one third of the median duration time (Miller et al., 2020), but no participants fell in this category. We also excluded five participants who reported that they understand Arabic language, two who self-identified as Muslim, and five participants who reported to be younger than 18 years.

Thus, the final sample consisted of 1006 participants. Among them, 464 identified as women, 541 as men, and 1 reported other gender identity. The mean age was 53.3 (SD = 15.8; range 18-81) and 38.47% participants self-identified as Christian, 57.26% as not religious, and 4.27% as of other religion. On a self-anchoring left-right scale, the distribution of political ideology was largely symmetric (3.28% strong left, 14.53% left, 12.54% center left, 24.78% center, 12.54% center right, 16.02% right, 1.89% strong right, 14.43% preferred not to answer, 1 undisclosed).

Design and Materials

We used a 2 (occurrence of tears: no tears vs. tears) x 2 (vocal expression: non-vocal vs. vocal) nested between-subjects experimental design. Vocal expression was nested in the tearful condition, thus resulting in three experimental conditions. Participants were asked to watch a brief video clip of a refugee woman (~30s), prepared using pieces of audiovisual

material from an interview with an asylum seeker from Syria, recorded by the non-governmental organization Refugee Today. We obtained formal approval to use the images for research purposes. Using the available video footage, we created the three conditions. In the non-vocal/non-tearful condition (n = 325), participants watched a video clip in which the target displayed a neutral facial expression, while being silent. In the non-vocal/tearful condition (n = 349), they watched a video clip in which the same target displays tears yet is silent as well. In the third vocal/tearful condition (n = 332), participants saw a video in which the same target displays tears while telling her personal story in Arabic language. In each video clip, the target was shown on the same plain background along with the same short narrative in which the real story of the asylum seeker from the original video is told. The narrative was presented in each video as subtitles in Dutch (see SOM for the content in English and Dutch).

Procedure

Participants were informed that the study examined the emotional and behavioral reactions to different types of images and stories of people in society and were randomly assigned to one of the three experimental conditions. They were instructed that they would be asked to watch a video clip with an asylum seeker who arrived from Syria to the Netherlands.[10] After giving informed consent, participants watched an embedded Youtube video set on autoplay, and without video controls (i.e., pause or accelerate the video) or possibility to continue with the survey until the video ends.[11] After watching the video, they filled in a questionnaire and were invited to donate money which they received for participation in this study to a real organization helping refugees (Refugee Today). At the

[10] Along the survey, we used a common Syrian female name when referring to the asylum seeker. Each participant was randomly shown one of ten selected names.

[11] Participants were able to re-watch the video if they needed to and we measured time they spent on this page of the survey.

end, participants were probed for suspicion, debriefed, and thanked for their participation in the study.

Measures

Participants filled in the following measures: emotions attributed to the target and participants' own emotions about the target's situation (in a randomized order), dependency- and autonomy-oriented helping intentions (in a randomized order), socio-demographic questions, and helping behavior, as well as some exploratory measures. We also included one question as an attention check in the survey.

Manipulation check. We measured to what extent the target was perceived as expressing sadness (3 items, "To what extent do you think [name] feels the following emotions? Sad/sorrowful/joyless?" $\alpha = .90$). For comparative purposes, we also assessed perceptions of expressed anger (3 items, "angry/offended/outraged" $\alpha = .88$), and fear (3 items, "afraid/scared/frightened," $\alpha = .89$).

Self-reported felt emotions. Participants were asked to what extent they felt different emotions about the situation [name] described in the video. Specifically, we assessed affective responses of compassion ("compassionate," "moved," and "touched," $\alpha = .94$, Batson et al., 1987), sadness ("sad," "sorrowful," "unhappy," $\alpha = .94$, Zentner et al., 2008), distress ("stressed," "troubled," "concerned," $\alpha = .90$, Batson et al., 1987), and moral anger ("angry," "offended," "outraged," $\alpha = .84$, Batson et al., 2007; 2009).

Dependency-oriented helping intentions. We captured dependency-oriented helping intentions (based on Becker et al., 2019) by asking participants on a slider scale from 1 (*very much unwilling*) to 100 (*very much willing*) to what extent they would be willing to help asylum seekers such as [name] who have arrived in the Netherlands with their immediate reception. Specifically, participants indicated to what extent they would be willing to support different initiatives by, for example, signing a petition or donating money (5 items, "Provide

basic supplies for asylum seekers like [name]", "Provide basic clothing and blankets to help asylum seekers like [name]," "Give asylum seekers like [name] temporary shelter," "Provide asylum seekers like [name] emotional support," "Ensure immediate medical assistance for asylum seekers like [name]", $\alpha = .94$).

Autonomy-oriented helping intentions. Based on Becker et al. (2019), participants were asked to what extent they would be willing to help asylum seekers such as [name] who have arrived in the Netherlands to "stand on their own two feet (to be self-reliant)" by supporting different initiatives by, for example signing a petition or donating money (6 items, "Grant asylum seekers like [name] full asylum in the Netherlands", "Prevent lengthy detentions of asylum seekers like [name] in camps or detention centers," "Provide free legal aid to asylum seekers like [name]," "Offer asylum seekers like [name] employment in their profession," "Guarantee their rights of political participation once they are granted asylum," "Free education in the Dutch language and society," $\alpha = .94$). We used the same slider scale as for dependency-oriented helping intentions.

Helping behavior (donation). After filling in questions on basic socio-demographic characteristics, participants were invited to donate points (or part of them) they earned from participation in this study to a non-profit organization Refugee Today. They were informed that the points represent real money that will be donated directly by the recruitment agency to Refugee Today which was described as "a web portal bringing focus on refugees like [name] by telling their situation and stories in sound and image and thus helping refugees". Since participants earned 100 points for participation in this study, they could choose not to donate at all (0 points) to up to 100 points ("Would you like to donate your points to support Refugee Today? If you do not want to donate to this initiative, please mark 0. Otherwise, please indicate the number of points you are willing to donate."). In order not to make participants feel pressured to donate points, they were told that this donation was completely

voluntary. Donations ranged from 0 to 100 points (M = 31.4, SD = 43.4). A total of 39.38% donated at least one point, while 25.15% donated all their payment.

Socio-demographic information. We also collected relevant socio-demographic information from participants, including gender, age, political orientation, the knowledge of Arabic language, and religious affiliation. For exploratory purposes, we also measured feelings of responsibility (2 items, "I feel it is my moral responsibility to help [name]" and "I feel that others are responsible for helping [name]", α = .59).

Analytical Strategy

We conducted all analyses in *R* (R Core Team, 2022), employing packages *laavan* (Rosseel, 2012), *dplyr* (version 1.0.9, Wickham et al., 2022), *janitor* (version 2.1.0, Firke, 2021), *effsize* (version 0.8.1, Torchiano, 2020), *rstatix* (version 0.7.0, Kassambara, 2021), *TOSTER* (Lakens, 2017), *bestNormalize* (Peterson & Cavanaugh, 2020), *PerformanceAnalytics* (version 2.0.4, Peterson & Carl, 2020), *corrr* (version 0.4.3, Kuhn et al., 2020), and data.table (version 1.14.2, Dowle & Srinivasan, 2021), among others. The full reproducible code is available at the OSF Project website. Across all studies, we set alpha at .05 and we used the conventional goodness-of-fit criteria to assess the model fit: Comparative Fit Index (*CFI*) and Tucker-Lewis Index (*TLI*) ≥ .90, and a Root-Mean-Square Error of Approximation (*RMSEA*) close to .06 (see Hu & Bentler, 1999).

The measurement model. We employed structural equation modeling with a maximum likelihood estimator. We tested a 6-factor measurement model: felt compassion, sadness, distress, moral anger, as well as dependency-oriented helping intentions and autonomy-oriented helping intentions. We additionally tested if the proposed 6-factor model had a better fit than any combination of 5 or 4 latent factors.[12] We additionally checked if the

[12] We also pre-registered the possibility of testing dependency-oriented and autonomy-oriented scales as formative latent factors. However, the reflective measurement model showed a good model fit, and the items

three types of emotions attributed to the target load on three separate dimensions: sadness, anger, and fear.

The confirmatory SEM model. To test the main hypotheses, we employed structural equation modeling (SEM) and estimated a model with one predictor (the occurrence of emotional tears: a contrast variable in which both tearful conditions will be assigned a value .33 and the non-tearful condition a value -.66), four latent factors as mediators (felt compassion, sadness, distress, and moral anger), and three outcome variables: dependency-oriented helping intentions, autonomy-oriented helping intentions (both latent factors), and donation behavior.[13] Given that the donation variable was not normally distributed, we transformed it using the bestNormalize package in R. We tested the effects of experimental manipulation on outcome variables also indirectly via the observer's emotions (latent mediators were correlated in the model). We applied bias-corrected bootstrapping with 10,000 replacement samples to obtain confidence intervals for the indirect effects.

The exploratory SEM analyses. We employed similar SEM analyses to explore several alternative models. First, we estimated all other possible indirect paths in the main confirmatory SEM model. Second, we tested a sequential model in which felt emotions were predictors of both dependency- and autonomy-oriented helping intentions, which both in turn predicted behavior. Third, we tested a sequential model in which the tears-helping link was explained sequentially via *perceived* sadness and *felt* compassion. Fourth, we explored the effects of tearful vocal vs. tearful non-vocal expression on both types of helping intentions and donation behavior via the four types of felt emotions. We created a contrast variable in which the vocal tearful condition was assigned a value .33, the non-vocal tearful condition a

composing both scales were highly intercorrelated. We therefore concluded that a formative model is not a plausible factor structure solution.

[13] We combined the vocal and non-vocal conditions in the main analyses since we did not have specific hypotheses for the role of the vocal component. However, we additionally tested the effects of each of the specific components in an exploratory manner, and presented these results in SOM.

value -.33 and the non-tearful condition a value 0. We also tested moderation effects by age, gender, political ideology, and perceived responsibility by estimating regression models with interaction terms and/or a multi-group SEM model when the moderator could be considered categorical (gender, political ideology).

Results

Manipulation Checks: Perceived Sadness, Fear, and Anger

Independent *t-tests* indicated that participants in the two tearful conditions, $M = 79.0$, $SD = 19.2$, perceived the target as expressing significantly more sadness than participants in the non-tearful condition, $M = 71.1$, $SD = 20.6$, $t(600) = 5.80$, $p < .001$, $d = 0.40$ [.26, .53].[14] We thus considered our manipulation successful, although the perceived sadness was relatively high in both conditions. We did not observe statistically significant differences between the tearful and non-tearful conditions in perceived anger, $t(655) = -0.53$, $p = .599$, $d = -0.04$ [-.16, .10]. Yet the target was perceived as expressing more fear in the tearful than in the non-tearful conditions, $t(625) = 3.81$, $p < .001$, $d = 0.26$ [.14, .39], although this effect was smaller compared to the effect on perceived sadness.[15]

The Measurement Model and Descriptive Findings

The main measurement model with six latent factors showed a good fit to the data and had a significantly better model fit than any of these alternative measurement models (see Table S3.1 in SOM). Mean scores, standard deviations and bivariate correlations in the total sample are shown in Table S4.1 in SOM. Violin plots for felt emotions and helping intentions and donation measures across the three conditions are shown in Figure 1.

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[14] Drawing on the sequential procedure for calculating the desired sample size (Lakens, 2014), we stopped the data collection after reaching the threshold of 150 participants and examined the data to check if our experimental manipulation worked properly. Given that the detected effect of tears on perceived sadness, $d = .27$ [-.07, .63], was not statistically smaller than $d = .50$ (i.e., at least *medium* in size), we continued the data collection. See SOM Section 2 for more details.

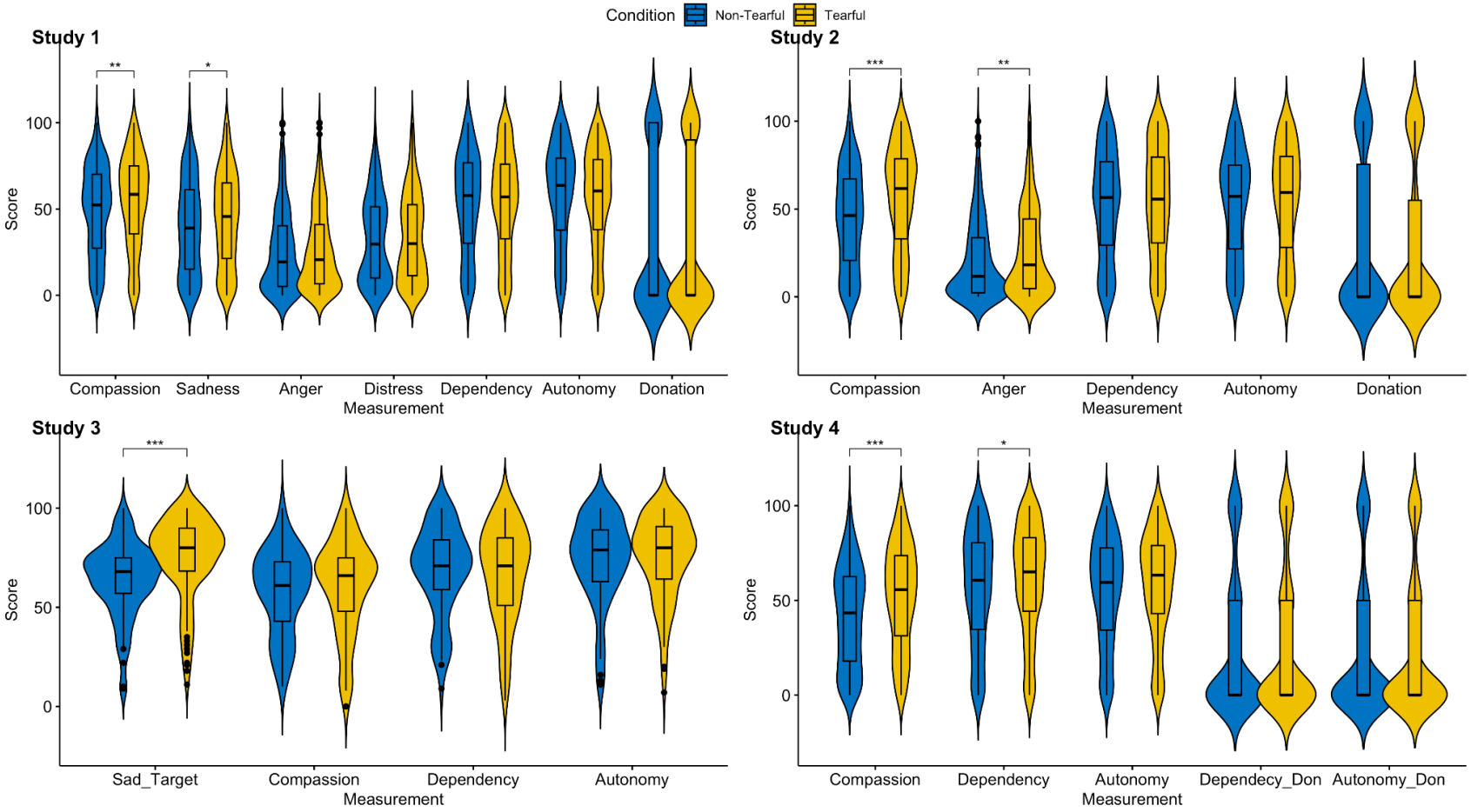
[15] In both online experiments, perceived sadness, fear, and anger loaded on three separate latent factors. See SOM Section 3.2 for details.

Confirmatory Model

Our main hypothesized structural model showed an acceptable fit, $\chi^2(253) = 2102.55$, $p < .001$, *CFI* = .92, *TLI* = .91, *RMSEA* = .09 [.08, .10] (see Table S5.1.1 in SOM for complete results). We compared this model to the full structural model in which all possible paths from emotions to helping intentions and donation were added. The full structural model showed significantly better fit to the data and represented the covariance matrix better, $\chi^2(253) = 1674.50$, $p < .001$, *CFI* = .94, *TLI* = .93, *RMSEA* = .08 [.07, .08], compared to the reduced, hypothesized model ($\Delta\chi^2(4) = 428.08$, $p < .001$). We therefore chose the full model as our final model (Figure 2 and Table S5.1.2).

Figure 1

*Violin plots for Felt Emotions and Intergroup Helping per Experimental Condition across the four studies. Statistically significant comparisons across conditions are depicted, * < .05, ** < .01, *** < .001.*

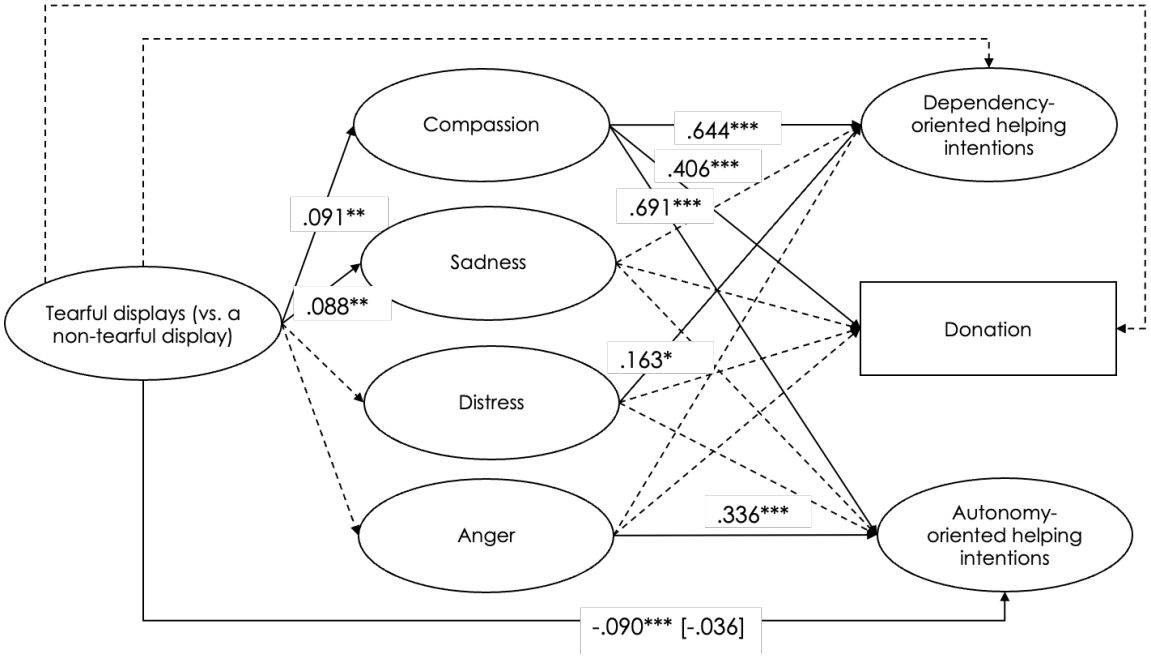


Results showed that, as expected, compared to the non-tearful display, tearful displays significantly increased observers' felt compassion and sadness (H1). However, we did not confirm our predictions regarding the effects of tearful expressions on distress and on moral anger (H1). We also did not observe a significant main effect of tearful (vs. non-tearful) expressions on dependency-oriented helping intentions, autonomy-oriented helping intentions or amount donated (H2—H3).

We further tested our hypotheses on indirect effects via emotional reactions (H4—H5). In the full structural model, we confirmed our hypothesis that the effect of tearful expressions on dependency-oriented helping intentions, $B = 3.65$, $SE = 1.34$, $p = .006$, 95%CI [1.12, 6.35], and donation, $B = 0.06$, $SE = 0.02$, $p = .012$, 95%CI [0.02, 0.11], was mediated by felt compassion, but not by self-oriented emotions such as sadness or distress. The expected indirect effect of tearful displays on autonomy-oriented helping intentions via increased moral anger was not statistically significant.

Figure 2

Confirmatory SEM Model: The Effects of Tears on Dependency-Oriented Helping Intention via Compassion, on Autonomy-Oriented Helping Intentions via Moral Anger, and Donation Behavior via Compassion, Distress, Sadness and Moral Anger (Study 1)



Note. N = 988. * p < .05, ** p < .01, *** p < .001. Model fit: χ^2 (253) = 2102.55, p < .001, *CFI* = .92, TLI = .91, *RMSEA* = .086 [.08, .10]. Correlations between the mediators and the outcomes are included but not shown for clarity. Standardized regression coefficients are shown. Dashed lines represented non-significant effects.

Exploratory Analyses

Additional Indirect Effects. In the same full main SEM model, we estimated additional indirect effects that were not hypothesized (see Table S5.1.2 in SOM). The analyses showed that the effect of tearful (vs non-tearful) displays on autonomy-oriented helping intentions was also explained by feeling compassion, B = 3.80, SE = 1.40, p = .006, 95%*CI* [1.14, 6.62].

Sequential Model: Tears-Felt Compassion-Intentions-Behavior. We tested a sequential model, with felt compassion as a mediator and in which donation was additionally regressed on dependency- and autonomy-oriented helping intentions. The model showed a good fit, χ^2 (96) = 750.39, p < .001, *CFI* = .96, *TLI* = .94, *RMSEA* = .083 [.08, .09]. There was a significant sequential indirect effect of tearful (vs non-tearful) displays on donation via felt compassion which subsequently predicted autonomy-oriented helping intentions, B =

0.04, *SE* = 0.02, *p* = .014, 95%*CI* [0.01, 0.08]. No similar effect was found via compassion and dependency-oriented helping intentions (Table S5.1.3 in SOM).

Sequential Model: Tears-Perceived Sadness-Felt Emotions-Helping Intentions & Behavior. We also estimated a sequential model in which tearful expression predicted helping intentions and behavior indirectly via perceived sadness and felt compassion as two sequential mediators. A SEM analysis indicated that the effects of the occurrence of tears had an impact on dependency-, *B* = 3.78, *SE* = 0.77, *p* < .001, 95%*CI* [2.35, 5.38], and autonomy-oriented helping intentions, *B* = 3.80, *SE* = 0.73, *p* < .001, 95%*CI* [2.45, 5.29], and donation, *B* = 0.06, *SE* = 0.02, *p* < .001, 95%*CI* [0.04, 0.10], indirectly and sequentially via perceived sadness and compassion. We did not observe sequential effects with perceived sadness and other felt emotions (Table S5.1.4 in SOM).

Effects of Tearful Non-vocal and Vocal Displays. In another exploratory model, we tested the effects of tearful vocal vs. non-vocal condition on outgroup helping, also indirectly via felt compassion (fit, χ^2 (264) = 1908.44, *p* < .001, *CFI* = .90, *TLI* = .88, *RMSEA* = .10 [.09, .10]). We did not observe significant effects of the vocal (vs. non-vocal) component on felt emotions, helping intentions and donation, or significant indirect effects (Table S5.1.5 in SOM).[16]

Moderation Effects. The findings were robust across women and men, right-wing, center, or left-wing political orientation, older and younger participants and those feeling lower and higher responsibility (see the summary of these analyses in SOM, Section 6.1).

Study 2

[16] The effects of tearful non-vocal and vocal expressions compared to the non-tearful expression separately replicated the indirect effects via compassion on helping intentions and behavior and are reported in SOM, Tables S5.1.6 (model fit: χ^2 (96) = 533.21, *p* < .001, *CFI* = .96, *TLI* = .94, *RMSEA* = .08 [.08, .09]) and S5.1.7 (model fit: χ^2 (96) = 517.89, *p* < .001, *CFI* = .96, *TLI* = .95, *RMSEA* = .08 [.08, .09]), respectively.

One possible reason for the lack of main (total) help-eliciting effects of emotional tears in Study 1 is that the perception of the target's sadness was high across both the tearful and non-tearful conditions. Participants might have focused on processing the sad narrative of the refugee woman rather than on the display of tears per se, and might have used the narrative to appraise emotions expressed in the video. As a result, the effects of tears on perceived sadness suffered from a ceiling effect caused by the emotional intensity of the story. To address this limitation, in Study 2 we exposed participants only to the two non-vocal (tearful vs. non-tearful) videos, and this time without the narrative presented in the subtitles. In addition, since in Study 1 our model was statistically complex and compassion and moral anger showed most consistent effects, in Study 2, we focused on these two emotions as processes involved in the link between tears and outgroup helping.

Method

Participants

As in Study 1, we calculated a desired sample size to obtain an effect size $d = 0.20$ when comparing two independent means, with an alpha of 0.05 and power of 0.8, and we thus aimed for 333 participants per experimental condition. We thus stopped data collection after reaching 670 participants who fully completed the survey.

The recruitment company Motivaction invited a different sample of native-born Dutch participants based on the same criteria as in Study 1. From an initial sample of 915 participants, we excluded 216 participants who did not pass the attention check (including 113 who failed the attention check question and 103 who did not complete the survey), and additional 4 participants who identified as Muslim and 3 who reported their age to be below 18. As in Study 1, no speeders were detected after the attention check filter (Miller et al., 2020), The final sample consisted of 692 participants (340 identified as men, 349 as women, and 3 reported other gender identity, age range: 18-87, $M_{age} = 52.8$, $SD = 15.2$). Most

participants self-identified as non-religious (55.78%), 41.18% as Christian, and 3.03% as of other religious identity. As for political orientation, 2.46% participants indicated strong left-wing orientation, 14.16% left, 16.18% center left, 23.70% center, 9.97% center right, 13.01% right, and 4.05% strong right-wing orientation (16.47% preferred not to answer).

Design and Materials

In Study 2, we relied on the two non-vocal conditions, the tearful (n = 350) and the non-tearful (n = 342) video clip but without a subtitled narrative. We thus used an experimental design in which only the occurrence of tears (no tears vs. tears) was manipulated. The procedure was the same as in Study 1.

Measures

All measures, including the ***manipulation checks:*** perceived sadness (α = .93), anger (α = .91) and fear (α = .90); ***felt emotions*** of compassion (α = .95) and moral anger (α = .91); ***dependency-oriented*** (α = .94), and ***autonomy-oriented helping intentions*** (α = .95), ***helping behavior (donation)***, and socio-demographic variables were the same as in Study 1.[17] Donated points ranged from 0 to 100 (M = 29.3, SD = 42.6). A total of 35.84% donated at least some part of their points and 23.12% donated their whole payment. Among the ***exploratory measures*** (feelings of responsibility, like in Study 2), in order to replicate the effects of tears on interpersonal social support observed in earlier studies (e.g., Zickfeld et al., 2021), we also assessed perceived social support with four items that formed two subdimensions: physical support ("I would give her a hug", "I would hold her hand", r = .85) and emotional support ("I would be there if she needed me," and "I would offer her support", r = .93). A confirmatory factor analysis showed that the two-factor model, χ^2 (6) =0.975, p < .001, CFI = 1.00, TLI = 1.00, $RMSEA$ = .000 [.000, .100], AIC = 24207.07, fit the data better

[17] In the scales measuring perceived and felt anger, the item "offended" was replaced by "indignant", given that in Study 2 participants did not report feeling "offended" compared to other items measuring anger.

than a one-factor model, χ^2 (6) =199.30, $p < .001$, *CFI* = .94, *TLI* = .81, *RMSEA* = .38 [.33, .42], *AIC* = 24403.40.

Results

The analytical models were identical to Study 1.

Manipulation Checks: Perceived Sadness, Fear, and Anger

As in Study 1, our experimental manipulation was successful but with a larger effect size. Specifically, participants in the tearful condition, $M = 83.5$, $SD = 18.0$, perceived the target as sadder than participants in the non-tearful condition, $M = 66.8$, $SD = 25.5$, $t(613) = 9.90$, $p < .001$, $d = 0.75$ [.61, .91].[18] We did not observe significant differences between the two conditions in perceived anger, $t(688) = 1.62$, $p = .106$, $d = 0.12$ [-.02, .27], yet similar to Study 1 the target was perceived as expressing more fear in the tearful than in the non-tearful condition, $t(685) = 2.89$, $p = .004$, $d = 0.22$ [.07, .37].

The Measurement Model and Descriptive Data

The 4-factor measurement model showed a good fit to the data, and a better fit than alternative models (Table S3.1 in SOM). Mean scores, standard deviations and bivariate correlations are shown in Table S4.2 in SOM. Violin plots for felt emotions and outgroup helping measures across the two conditions are shown in Figure 1.

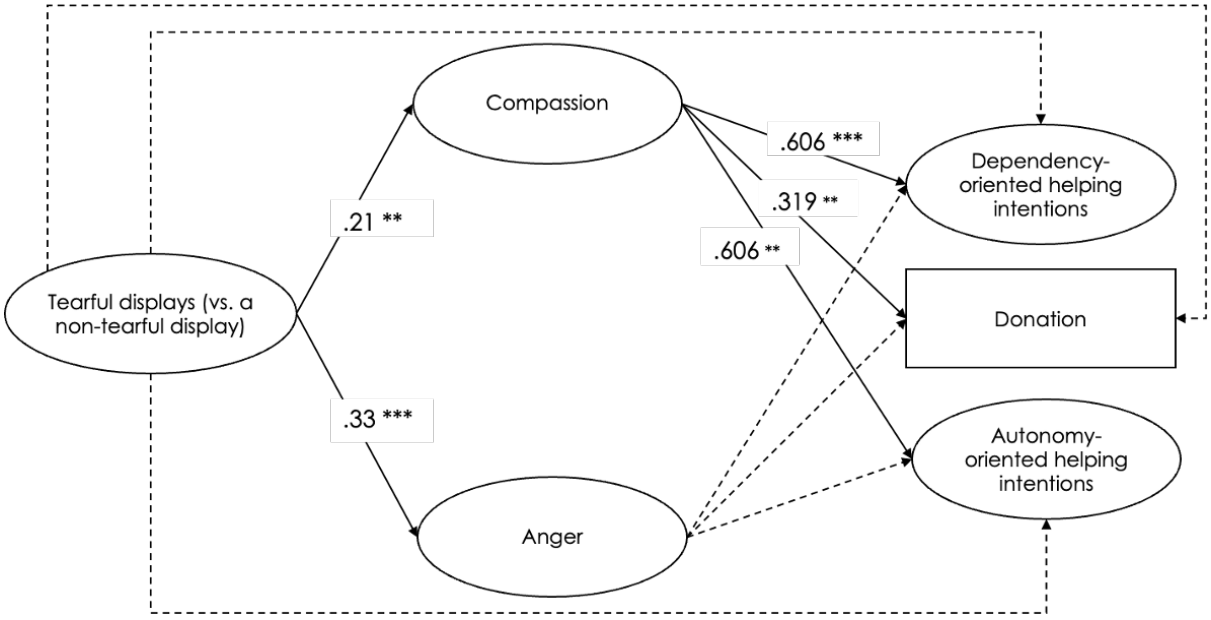
Confirmatory Model

[18]Again, we stopped the data collection after reaching a threshold of 50 participants per condition to check if our experimental manipulation worked properly (Lakens, 2014). Based on previous research (Bobowik et al., 2022; Zickfeld et al., 2021), we assumed that in Study 2 without the narrative, the effect of tearful displays compared to the non-tearful display on perceived sadness should not be smaller than $d = .86$. As in Study 1, the equivalence test, that is, the difference between the obtained effect ($d = .94$) and the benchmark ($d = .86$) was non-significant, $t(95.8) = 0.385$, $p = 0.649$, so we decided to continue the data collection. See SOM Section 2 for details.

The main model showed a good fit to the data, χ^2 (139) =777.39, $p < .001$, *CFI* = .95, *TLI* = .94, *RMSEA* = .08 [.08, .09] (Figure 3). A tearful display increased participants' felt compassion and moral anger compared to the non-tearful display, corroborating our predictions (H1). As in Study 1, we did not find a significant main (i.e., total) effect of tearful (vs. non-tearful) display on dependency-oriented helping intentions, autonomy-oriented helping intentions or amount donated to a real initiative oriented at helping refugees (H2-H3). Yet, tears had effects on dependency-oriented, *B* = 7.55, *SE* = 1.49, $p < .001$, 95%*CI* [4.71, 10.59], and autonomy-oriented helping intentions, *B* = 7.47, *SE* = 1.42, $p < .001$, 95%*CI* [4.75, 10.35], and donation, *B* = 0.09, *SE* = 0.02, $p < .001$, 95%*CI* [0.06, 0.14] indirectly via feelings of compassion (H4 and H6). We did not find significant indirect effects of emotional tears via moral anger (H5). Complete analyses are shown in Table S5.2.1 in SOM.

Figure 3

Confirmatory SEM Model: The Effects of Tears on Dependency- and Autonomy-Oriented Helping Intentions and Helping Behavior via Compassion and Anger (Study 2)



Note. *N* = 688. * $p < .05$, ** $p < .01$, *** $p < .001$. Model fit: χ^2 (139) = 777.39, $p < .001$, *CFI* = .95, *TLI* = .94, *RMSEA* = .08 [.08, .09]. Correlations between the mediators and the outcomes are included but not shown for clarity. Standardized regression coefficients are shown. Dashed lines represented non-significant effects.

Exploratory Analyses

Sequential Model: Tears-Felt Emotions-Intentions-Behavior. We again also estimated a sequential model, in which donation was regressed on dependency- and autonomy-oriented helping intentions. The model showed a good fit, χ^2 (139) = 777.40, $p <$ .001, *CFI* = .95, *TLI* = .94, *RMSEA* = .08 [.08, .09]. As in Study 1, there was a significant sequential indirect effect of tearful (vs non-tearful) displays on donation via compassion which subsequently predicted autonomy-oriented helping intentions, *B* = 0.07, *SE* = 0.02, $p <$ .001, 95%*CI* [0.04, 0.12]. No similar effect was found via compassion and dependency-oriented helping intentions (see Table S5.2.2 in SOM).

Sequential Model: Tears-Perceived Sadness-Felt Emotions-Helping Intentions & Behavior. We also estimated a sequential model in which tears influenced helping intentions and behavior indirectly and sequentially via *perceived* sadness and *felt* compassion. A SEM analysis showed that the effects of the occurrence of tears had an impact on dependency-oriented, *B* = 7.01, *SE* = 1.08, $p <$ .001, 95%*CI* [5.02, 9.28], and autonomy-oriented helping intentions, *B* = 6.87, *SE* = 1.03, $p <$ .001, 95%*CI* [4.96, 9.03], and donation behavior, *B* = 0.09, *SE* = 0.02, $p <$ .001, 95%*CI* [0.05, 0.13] indirectly and sequentially via perceived sadness and felt compassion (Table S5.2.3 in SOM). We did not observe sequential effects with perceived sadness and moral anger.

The Effects of Tears on Social Support Intentions. We additionally tested a similar model in which the dependent variable was social support intention. We fitted a model with two dependent variables: intentions to offer physical support (i.e, hold a hand or give a hug) and intentions to offer emotional support (i.e., be there for someone). This model showed good fit to the data, $\chi^2(35) = 113.15$, $p <$ .001, *CFI* = .989, *TLI* = .983, *RMSEA* = .057, 90% *CI* [0.045, 0.069], and indicated that people are more willing to offer physical support when the refugee woman is displaying tears, whereas no similar effect was observed for emotional

support intentions. In addition, the link between tearful displays and willingness to offer physical support, $B = 8.19$, $SE = 1.59$, $p < .001$, 95%CI [5.07, 11.37], and emotional support was explained by feelings of compassion, $B = 8.81$, $SE = 1.71$, $p < .001$, 95%CI [5.46, 12.17] (Table S5.2.4 in SOM).

Moderation Effects. The findings were again robust across men and women, right-wing, center, or left-wing political orientation, older and younger participants and those feeling lower and higher responsibility (see SOM, Section 6.2).

Study 3

Study 3 was a field study, in which we implemented an exhibition-based quasi-experiment, testing the effects of the expression of tears on helping intentions and behavior outside the laboratory. The study was a citizen science initiative bridging visual journalism, photography, social science, and fundraising action. Besides ourselves, several parties were involved in the design of the study, including the university public engagement team, photographers, graphic designers, and event producers. Participants in this study were visitors of "Betweter Festival", a festival for science, arts, music, film, and live experiments organized by the local university. In this study, we exposed participants to either a tearful or a non-tearful version of a set of portraits of refugees and measured feelings of compassion, dependency- and autonomy-oriented helping intentions, and the amount donated to a social initiative supporting refugees. We focused on feelings of compassion only as this showed the most consistent effects across Studies 1 and 2, and we could not include too many measures due to practical constraints.

Method

Participants

Participants were recruited among the visitors of the science festival. The sample consisted of 209 participants. Although we calculated a priori a larger desired sample size

(see SOM for details), we recruited as many participants as possible during the event. As in Studies 1 and 2, we excluded participants younger than eighteen years-of-age (n = 2) and sixteen participants did not finish the survey. Hence, the final sample size was 191 and a sensitivity analysis showed that this provided 80% power (α = .05, one-tailed) to detect an effect size of d = 37. In this sample, 62 identified as men, 122 as women, 4 as non-binary, and 3 reported other gender identity. The mean age was 34.2 (SD = 12.4), and age ranged from 18 to 73. Most participants (n = 172) were born in the Netherlands, but 19 were foreign-born.

Design and Materials

The study was carried out in collaboration with the non-governmental organization Refugee Today. The study included a quasi-experimental design with one between-subject factor (tearful, n = 118, vs. non-tearful condition, n = 73). Each of the two conditions included 10 portrait high-resolution prints of the same protagonists (five women, five men, 70 x 96cm) taken by a photographer from Refugee Today. The ten portraits were selected based on a prior validation study of 26 images that were digitally manipulated to add visible tears (see Section 1.3 in SOM for a detailed description of the validation phase).

Procedure

Participants were recruited on September 30th, 2022, among the visitors of the "Betweter Festival" at Tivoli Vredenburg, Utrecht. The festival started at 6:30pm, and the data collection started with the first participants, at 7pm. We then alternated the two conditions every hour throughout the duration of the science festival, till 12pm, starting with the tearful condition.[19]

[19] Among the 209 total participants, the following number of participants visited the exhibition and filled in a survey (or at least started it) each hour: 19:00-20:00 - 59 participants; 20:00-21:00 - 11 participants; 21:00-22:00 - 37 participants; 22:00-23:00 - 70 participants; 23:00-00:00 - 32 participants. Although the festival was programmed till 01:00, the experiments ended by 00:00.

Before participating in the experiment, participants read information about the study placed on an LCD screen at the entrance to the exhibition and were invited to visit the exhibition and participate in the study by two confederates. After visiting the exhibition, participants saw a panel with information about the purpose of the exhibition and Refugee Today. They were asked to read instructions and to scan a QR code with a link to a brief Qualtrics survey about their feelings and perceptions regarding the portraits. Two additional confederates assisted participants at this point and reminded them about the survey being accessible via a QR code on a panel placed at the exit of the exhibition. At the end of the survey participants were offered an opportunity to donate money, and if they chose to do so, they were redirected to a website, where they could make an online donation.

Measures

Given time constraints, participants were invited to respond to a brief survey with single-item measures, including a ***manipulation check*** ("To what extent do you think the refugees shown in the exhibition express sadness?"), ***felt compassion*** ("When visiting the exhibition, to what extent have you felt touched?"), ***dependency-*** ("To what extent would you be willing to help refugees, like the ones shown in the exhibition, with their immediate reception (e.g., by donating clothing or food)?") and ***autonomy-oriented helping intentions*** ("To what extent would you be willing to help refugees, like the ones shown in the exhibition, to stand on their own two feet and to be self-reliant (e.g., by signing a petition to guarantee their employment or rights of political participation)?"). The responses for these measures were placed on a slider-scale ranging from 0 (*not at all*) to 100 (*very much*). We also collected relevant socio-demographic information (i.e., age, gender, being Dutch- vs. foreign-born), and measured ***prosocial behavior*** by inviting participants to take part in a real fundraising action to install an exhibition with portraits of Ukrainian refugees in Kyiv. A total of 22 participants (11.6%) donated at least €1, and the donations ranged from €1 to €25 (M =

7.91, *SD* = 5.71). We also controlled if participants passed by the exhibition corridor earlier (146 reported not having passed by earlier, 46 have passed by earlier).

Analytical Strategy

To test the main hypotheses, we estimated a path model with the occurrence of emotional tears (coded as 0 and 1) as predictor, felt compassion as mediator, and three outcome variables: dependency- oriented helping intentions, autonomy-oriented helping intentions, and donation behavior. The donation variable was transformed as in Studies 1 and 2.

Results

Manipulation Check: Perceived Sadness

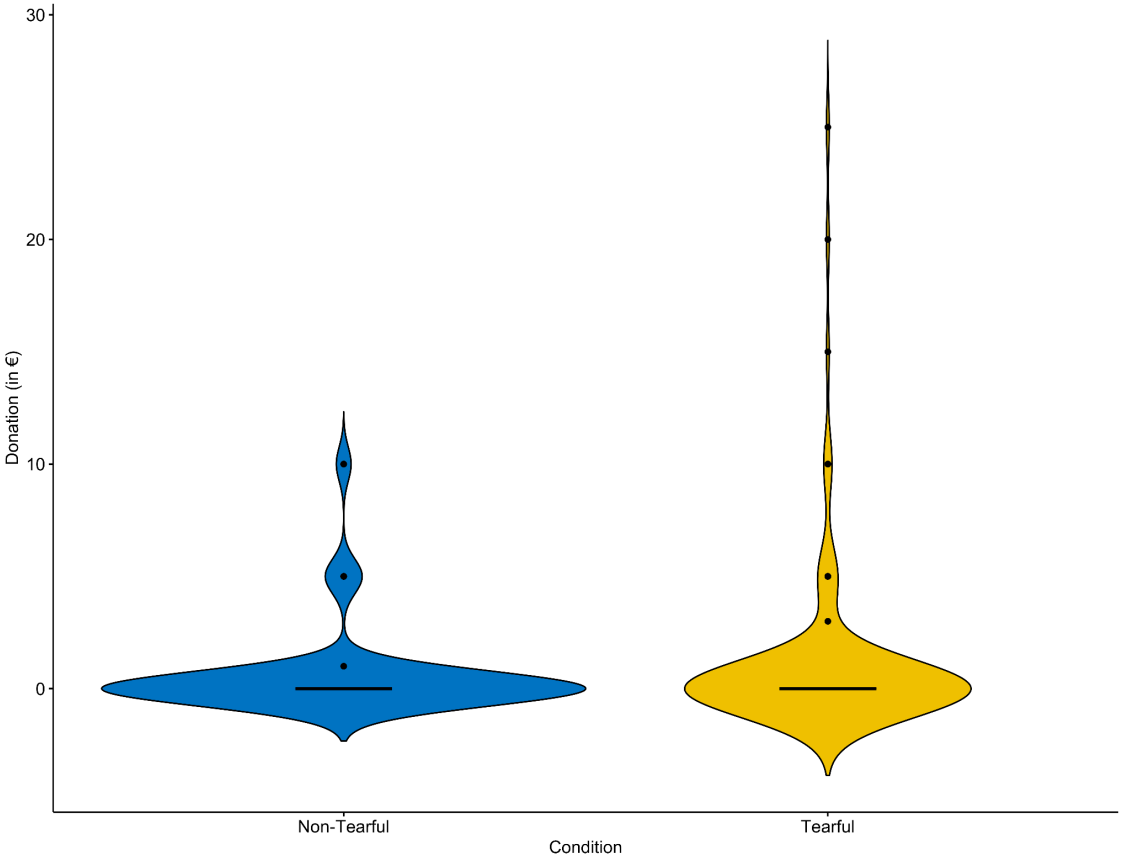
Participants in the tearful condition (*M* = 74.9, *SD* = 20.5) perceived the target as expressing significantly more sadness than participants in the non-tearful condition (*M* = 64.1, *SD* = 18.7), $t(163) = 3.74$, $p < .001$, $d = 0.55$ [.26, .87].[20]

Descriptive Data

Mean scores, standard deviations and bivariate correlations in the total sample are shown in Table S4.3 in SOM. Violin plots for the main variables per condition are shown in Figure 1 and 4.

Figure 4
Violin Plot for Helping Behavior (b) per Experimental Condition (Study 3)

[20] Robustness checks showed that this effect was similar in strength after excluding foreigners, $d = 0.58$ [.25, .90], and those who passed by the exhibition earlier, $d = 0.44$ [.12, .78].



Confirmatory Model

A path analysis did not reveal significant effects of tearful (vs. non-tearful) expressions on felt compassion (H1), dependency-oriented helping intentions, autonomy-oriented helping intentions or amount donated (H2-H3). Indirect effects were not significant either (Table S5.3.1 in SOM for detailed analyses). The results held the same when we controlled the analyses for being born in the Netherlands and having passed by the exhibition earlier (Table S5.3.2 in SOM).

Exploratory Analyses[21]

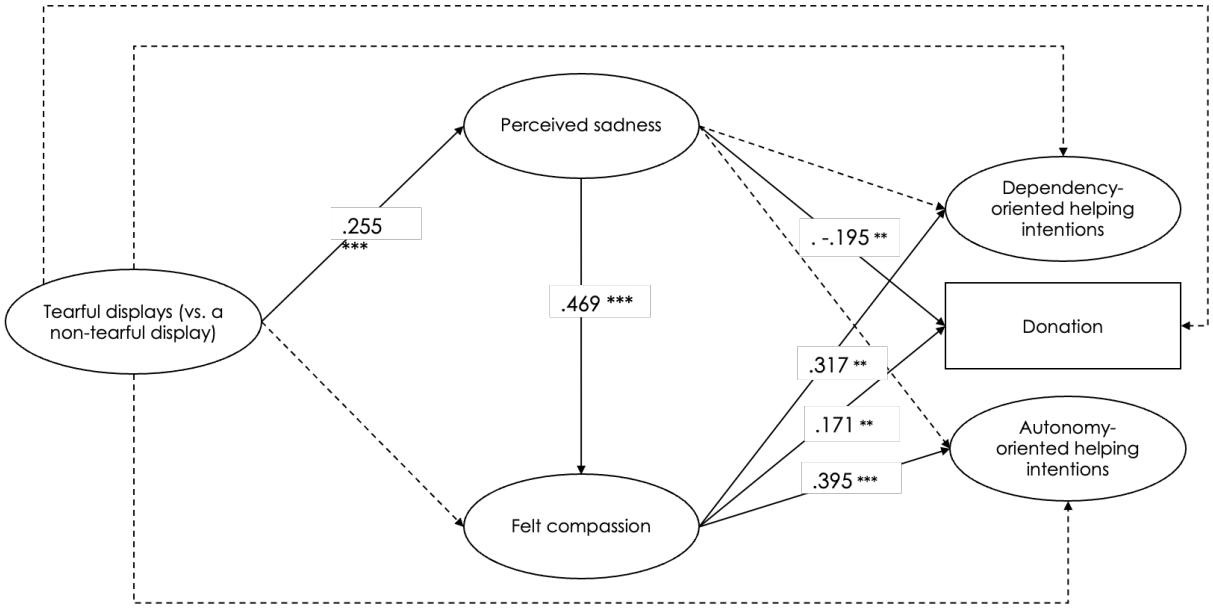
Sequential Model: Tears-Perceived Sadness-Felt Emotions-Helping Intentions & Behavior. Again, we estimated a sequential model in which tears influenced helping

[21] Given a non-significant effect of occurrence of tears on felt compassion, we did not test a sequential model in which compassion predicts dependency- and autonomy-oriented helping intentions, and helping intentions are predictive of helping behavior.

intentions and behavior indirectly and sequentially via perceived sadness of the refugee targets and personally felt compassion. An (identified) path analysis showed such a sequential indirect effect of tears on dependency- (B = 1.84, SE = 0.76, p = .015, 95%CI [0.56, 3.55]) and autonomy-oriented helping intentions (B = 2.07, SE = 0.78, p = .008, 95%CI [0.70, 3.74]) as well as helping behavior (B = 0.04, SE = 0.02, p = .037, 95%CI [0.01, 0.09], see Figure 5). Complete path analysis results with and without covariates are available in SOM, Table S5.3.3 and Table S5.3.4, respectively.

Figure 5

Exploratory Path Model: The Effects of Tears on Dependency-Oriented Helping Intentions, Autonomy-Oriented Helping Intentions, and Donation Behavior Sequentially via Perceived Sadness and Compassion (Study 3)



Note. N = 193. * p < .05, ** p < .01, *** p < .001. Standardized regression coefficients are shown. Dashed lines represent non-significant effects.

Moderation Effects. We did not observe significant interaction effects between condition and gender or condition and age, except for a significant interaction between condition and age in explaining autonomy-oriented helping intentions (see SOM, Section 6.3).

Study 4

Study 4 was an online experiment conducted with a large, nation-wide sample of Dutch participants, similar to Studies 1 and 2. The objective of this study was to replicate the indirect link between tears and help via feelings of compassion, considering two types of helping intentions (i.e., willingness to offer dependency-oriented and autonomy-oriented help) and corresponding behaviors (donations to either a dependency-oriented or autonomy-oriented initiative). Additionally, the study aimed to investigate the impact of tearful portrayals of refugees on participants' perception of refugees' needs.[22] Unlike Studies 1, 2, and 3, which used videos featuring a single expresser woman or a set of ten portraits exhibited simultaneously, in Study 4, participants were randomly exposed to one of multiple images of tearful and non-tearful refugee men and women. By including both gender categories in the images, we ensured that any observed effects were not driven by the gender of the portrayed individuals. Second, by varying the refugees portrayed in the images, the study avoided potential confounds associated with specific individual characteristics.

Method

Participants

Based on the non-significant main effects observed in Studies 1-3, we focused on the average sequential indirect effect obtained from this data to determine the appropriate sample size ($\beta = .0174$). We conducted a power analysis using a ShinyApp for power in mediation analyses (Schoemann et al., 2017). We varied the overall sample size from 200 to 1000 participants, with alpha set at .05. The analysis revealed that a power of 95% could be achieved for the sequential mediation effect (a1db2) with a sample size of 850 participants. Considering the possibility of participant exclusions, we aimed to collect data from a total of

[22] The results did not yield significant effects of experimental manipulation on perceived needs.

900 participants, with 450 participants assigned to each condition. Data collection was concluded once the target of 949 fully completed survey responses was reached.

Native-born Dutch participants were recruited for the study through a research company, Motivaction, using the same criteria as in Studies 1 and 2. Initially, a total of 1225 participants were included in the study. From this initial sample, we excluded participants for various reasons: 185 participants failed the attention check question, 81 participants did not complete the survey, and 6 participants reported their age to be below 18 (and one did not provide their age). Three speeders were identified and excluded following the attention check filter, similar to the findings in Study 1 (Miller et al., 2020). The final sample for analysis consisted of 949 participants, including 553 who identified as men, 392 as women, and 4 participants who reported other gender identities. The age range of the participants was 18-80 years, with a mean age of 50.2 (SD = 15.6). Regarding political orientation, 2.95% of participants indicated a strong left-wing orientation, 12.22% identified as left-leaning, 11.38% as center-left, 21.60% as center, 14.12% as center-right, 13.17% as right-leaning, and 3.27% as having a strong right-wing orientation. Additionally, 21.29% of participants preferred not to answer the question regarding political orientation.

Design and Materials

We used a between-subject experimental design in which participants were randomly assigned to either a non-tearful condition (n = 444) or a tearful condition (n = 505) and thus exposed to a photograph of a refugee either without tears or with visible tears digitally added to the photograph. Each of the two conditions included 10 portrait high-resolution prints of the same protagonists (five women, five men, 70 x 96cm) taken by a photographer from Refugee Today. The ten portraits were selected based on a prior validation study of 26 images that were digitally manipulated to add visible tears (see Section 1.3 in SOM for a detailed description of the validation phase).

Procedure

Procedure was similar as in Studies 1 and 2. Participants were randomly assigned to one of the two experimental conditions in which they observed one of the portraits of refugees. After observing the image, they filled in a questionnaire and were invited to donate money which they received for participation in this study to a real organization helping refugees (UNHCR). At the end, participants were probed for suspicion, debriefed, and thanked for their participation in the study.

Measures

The measures, including the ***manipulation check*** (i.e., perceived sadness***,*** α = .94), ***felt compassion*** (α = .94), ***dependency-oriented*** (α = . 96), and ***autonomy-oriented helping intentions*** (α = . 91), and were the same as in Study 1. In the case of DOHI and AOHI, we decided to retain three items per each dimension. In both dimensions, we dropped two items that were less clearly reflecting dependency-oriented help ("Give asylum seekers temporary shelter" and "Provide asylum seekers emotional support") or autonomy-oriented help ("Grant asylum seekers full asylum in the Netherlands", "Prevent lengthy detentions of asylum seekers in camps or detention centers," and "Provide free legal aid to asylum seekers"). With the aim of measuring two types of ***helping behavior (donation)*** participants were invited to donate points (100 points in total) offered for participation in this study, to a real organization helping refugees (specifically, UNHCR). They were invited to distribute their donation across two different real initiatives launched by the organization, a dependency-oriented and an autonomy-oriented initiative: "We invite you to donate money to one or both initiatives launched by the UNHCR. You can distribute 100 points you earned as you wish. If you choose to donate, these points will be donated directly by Motivaction to UNHCR: a) To help refugees, like the ones shown in the photos, by providing basic supplies, such as warm clothing, food, shelter, or fuel (for instance, to survive this winter). Would you like to donate

money to this initiative?; b) To help refugees, like the ones shown in the photos, to stand on their own two feet and to be self-reliant (e.g., by guaranteeing their fast employment or free access to education). Would you like to donate money to this initiative?" For both types of help, donated points ranged from 0 to 100 (Autonomy-oriented initiative: $M = 22.8$, $SD = 35.6$; Dependency-oriented initiative: $M = 25.9$, $SD = 37.0$). A total of 39.30% donated at least some part of their points and 12.85% donated their whole payment to the autonomy-oriented initiative. A total of 42.26% donated at least some part of their points and 14.12% donated their whole payment to the autonomy-oriented initiative. The study also included measures of ***perceived needs*** (see SOM for more details).

Analytical Strategy

The analytical strategy was similar as in Studies 1-3. However, considering that we measured two types of helping intentions and two types of helping behavior, we directly tested a sequential model, in which the occurrence of emotional tears (coded as 0 and 1) was a predictor variable, felt compassion was the first mediator, the dependency-oriented and autonomy-oriented helping intentions were two parallel mediators regressed on felt compassion, and dependency-oriented and autonomy-oriented behavior were outcome variables. The helping behavior variables were transformed as in Studies 1 and 2.

Results

Manipulation Checks: Perceived Sadness

As in Studies 1-3, our experimental manipulation was successful. Participants in the tearful condition, $M = 74.2$, $SD = 22.7$, perceived the target as sadder than participants in the non-tearful condition, $M = 57.9$, $SD = 25.1$, $t(900) = 10.4$, $p < .001$, $d = 0.68$ [.56, .81].

The Measurement Model and Descriptive Data

The 3-factor measurement model showed excellent fit to the data, and a better fit than alternative models (Table S3.1 in SOM). Mean scores, standard deviations and bivariate

correlations are shown in Table S4.2 in SOM. Violin plots for felt compassion and helping measures across the two conditions are shown in Figure 1.

Confirmatory Model

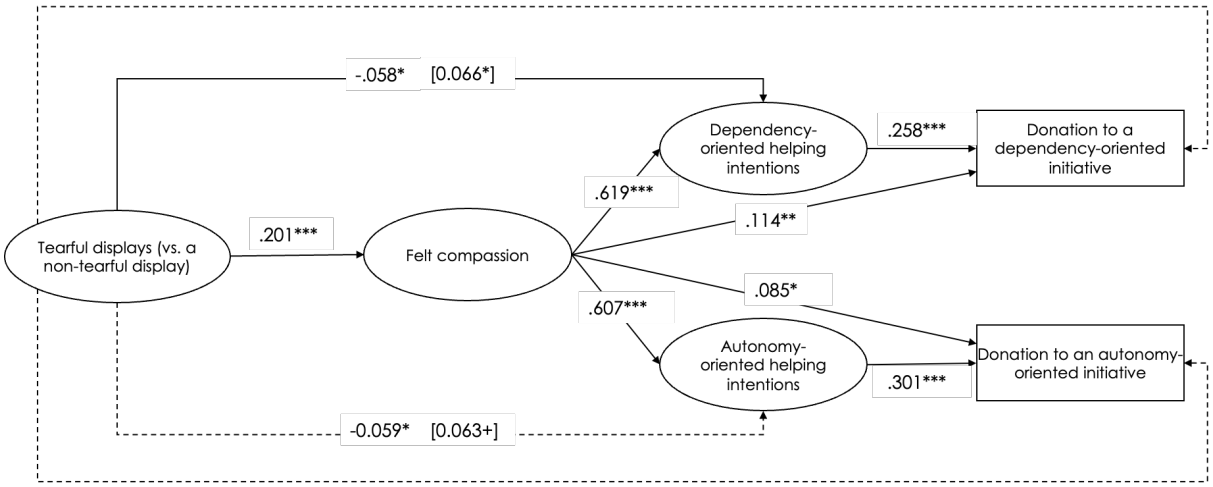
The main model showed a good fit to the data, χ^2 (44) = 129.719, $p < .001$, *CFI* = .99,2 *TLI* = .988, *RMSEA* = .045 [.036, .055] (Figure 6). A tearful display increased participants' felt compassion compared to the non-tearful display, corroborating our predictions (H1). As in Studies 1-3, we did not find a significant main effect of tearful (vs. non-tearful) display on autonomy-oriented helping intentions or amount donated to any of the two types of initiatives oriented at helping refugees (H2-H3), but there was a significant effect on dependency-oriented helping intentions.

Supporting H4 and H6, tears were associated with dependency-oriented intentions, *B* = 0.31, *SE* = 0.06, $p < .001$, 95%*CI* [0.21, 0.43], and behavior, *B* = 0.04, *SE* = 0.01, $p = .013$, 95%*CI* [0.01, 0.07], as well as with autonomy-oriented helping intentions, *B* = 0.301, *SE* = 0.06, $p < .001$, 95%*CI* [0.20, 0.42], and behavior, *B* = 0.03, *SE* = 0.01, $p = .057$, 95%*CI* [0.001, 0.05], indirectly via feelings of compassion.

We additionally observed significant sequential indirect effects, corroborating H 7 and H8. The displays of refugees tears led to more feelings of compassion, which in turn were linked with more dependency-oriented intentions and in consequence behavior (donation to an UNHCR initiative helping refugees to survive winter), *B* = 0.05, *SE* = 0.01, $p < .001$, %*CI* [0.03, 0.07] . In addition, th displays of refugees tears led to more feelings of compassion, which in turn were linked with more autonomy-oriented intentions and in consequence behavior (donation to an UNHCR initiative giving refugees education opportunities), *B* = 0.06, *SE* = 0.01, $p < .001$, 95%*CI* [0.03, 0.08]. Complete analyses are shown in Table S5.4.1 in SOM.

Figure 6

Confirmatory Sequential SEM Model: The Effects of Tears on Dependency- and Autonomy-Oriented Helping Intentions and Helping Behavior via Compassion (Study 4)



Note. $N = 949$. * $p < .05$, ** $p < .01$, *** $p < .001$. Model fit: χ^2 (44) =129.72, $p < .001$, *CFI* = .99, *TLI* = .99, *RMSEA* = .05 [.04, .06]. Correlations between the mediators and the outcomes are included but not shown for clarity. Standardized regression coefficients are shown. Dashed lines represented non-significant effects.

Exploratory Analyses

Sequential Model: Tears-Perceived Sadness-Felt Compassion-Helping Behavior. We also estimated a sequential model in which tears predicted two types of helping behavior indirectly and sequentially via *perceived* sadness and *felt* compassion (model fit: $\chi^2(20) = 123.35$, $p < .001$, *CFI* = .986, *TLI* = .975, *RMSEA* = .074, 90% *CI* [0.062, 0.087]). A SEM analysis showed that the effects of the occurrence of tears had an impact on dependency-oriented, $B = 0.09$, $SE = 0.02$, $p < .001$, 95%*CI* [0.05, 0.12], and autonomy-oriented helping behavior, $B = 0.08$, $SE = 0.02$, $p < .001$, 95%*CI* [0.05, 0.11], indirectly and sequentially via perceived sadness and compassion (Table S5.4.2 in the SOM).

Moderation Effects. As in Studies 1-3, The findings were robust across participants who identified as men and women, with right-wing, center, or left-wing political orientation, as well as older and younger participants (see the summary of these analyses in the SOM,

Section 6.1). We additionally tested if the target's gender moderated the effects. There were no significant moderation effects by the target's gender.

Meta-Analytical Integration

Following the criteria of mini meta-analysis in the literature (e.g. Goh et al., 2016), we conducted a meta-analytical integration of the results from Studies 1-4 to obtain more robust estimations of the examined effects (Table 1). Forest plots are shown in Figures S7.1—14 in SOM (for more information on the meta-models please see SOM Section 7). The meta-analytical integration of the data showed an overall significant effect of tearful (vs. non-tearful) expressions on perceptions of refugees' sadness and fear, as well as participants' own felt compassion. In addition, we found an overall significant positive correlation between compassion and dependency- and autonomy-oriented helping intentions and donation amount. Finally, we observed significant average indirect effects of tears on dependency- and autonomy-oriented helping intentions via feelings of compassion, whereas this effect was not statistically significant in the case of the donation, though positive and comparable in strength.

Table 1

Meta-Analytical Integration of Results from Studies 1-4

Effect	Esti-mate	95% CI	*p*	*Q*	*df*	p_Q	I^2	95%*CI* (I^2)	
Std. Mean Differences – Condition (-.5: Non-Tearful, .5: Tearful) (Estimate = *d*)									
Perceived sadness	0.60	[0.43, 0.77]	< .001	14.26	3	0.00	76.74	28.21	98.07
Perceived fear	0.24	[0.15, 0.34]	< .001	0.18	1	0.67	0.00	0.00	99.46
Perceived anger	0.04	[-0.12, 0.20]	.623	2.48	1	0.12	59.62	0.00	99.96
Felt compassion	0.30	[0.17, 0.44]	< .001	8.50	3	0.04	64.87	0.00	98.06
Felt anger	0.12	[-0.04, 0.29]	.149	2.79	1	0.09	64.17	0.00	99.96
DOHI	0.07	[-0.00, 0.15]	.065	1.90	3	0.59	0.00	0.00	91.88
AOHI	0.04	[-0.06, 0.14]	.410	4.13	3	0.25	36.04	0.00	92.79
Donation behavior	0.02	[-0.06, 0.09]	.690	1.18	3	0.76	0.00	0.00	90.47
Correlations Compassion – Helping (Estimate = *r*)									
DOHI	0.53	[0.43, 0.64]	< .001	17.05	3	0.00	93.69	73.79	99.67
AOHI	0.55	[0.44, 0.65]	< .001	21.01	3	0.00	93.85	75.30	99.67
Donation behavior	0.24	[0.21, 0.28]	< .001	4.97	3	0.17	0.07	0.00	98.54
Indirect Effects – X (Condition) -> M (Compassion) -> Y (Helping) (Estimate = β)									
DOHI	0.08	[0.03, 0.12]	.001	14.80	3	0.00	80.94	36.97	98.69
AOHI	0.08	[0.03, 0.13]	.001	13.60	3	0.00	79.68	34.75	98.64
Donation behavior	0.03	[0.01, 0.06]	.008	21.12	3	0.00	85.57	56.25	98.96

Note. *Q*(df) = k-1. DOHI = Dependency-Oriented Helping Intentions; AOHI = Autonomy-Oriented Helping Intentions.

Further, heterogeneity tests (i.e., *Q*, I^2) showed that the main effects of tears were all (relatively) homogeneous except for the effects on attributions of sadness and felt compassion. Conversely, the correlations between compassion and helping intentions and donation, as well as indirect effects were heterogeneous across studies.

General Discussion

In three experiments relying on nation-wide panel data and a field, exhibition-based quasi-experiment, we assessed the impact of tearful images of refugees on eliciting help. We measured intentions to offer two types of assistance to refugees - dependency- and autonomy-oriented help - and actual help provided. We also explored the role of emotional reactions as explanatory mechanisms. The findings across these studies indicated that depictions of refugees showing emotional tears are predictive of people's willingness to offer refugees both immediate (dependency-oriented) and more agency-promoting (i.e., autonomy-oriented) help, as well as actual meaningful helping behaviors, via compassionate feelings activated in response to these emotive images. The effects of exposure to tearful refugee images on the amount donated to an organization supporting refugees' empowerment occurred also indirectly and sequentially, through pathways of compassion and intentions to help promote refugees' autonomy. Study 4 additionally showed that this sequential process (via compassion and dependency-oriented helping intentions) applies to dependency-oriented helping initiatives. In addition, the exposure to tearful portrayals of refugees translated into more helping intentions and behavior indirectly via perceived sadness of the target(s), which activated more feelings of compassion toward the refugee(s).

The Impact of Tears on Felt Emotions towards Refugees

A key finding from the meta-analytical integration of data is the heightened sense of compassion members of the receiving society feel when exposed to images of refugees shedding tears. This corroborates existing evidence on the role of tears in evoking empathic feelings (e.g., Zickfeld et al, 2021), also in interethnic contexts (e.g., Bobowik et al., 2022). This effect, however, was more pronounced in Studies 2 and 4 compared to Studies 1 and 3. This variance could be attributed to the differences in perceived sadness of the targets, as indicated by the experimental manipulation checks ($d = 0.40$ in Study 1 and $d = 0.55$ in Study 3 vs. $d = 0.75$ in Study 2 and $d = 0.68$ in Study 4). These outcomes could provide insights

into the ongoing debate concerning the fluctuating levels of compassion or 'compassion fatigue' observed in response to media portrayals of arriving refugees (Kędra & Sommier, 2018). Some scholars advocate for a shift in media representation, suggesting that featuring asylum seekers with distinguishable facial features could foster more compassionate responses (Bleiker et al., 2013). In our research, the visual representations of refugees diverged from the depersonalized mass images commonly seen in media (e.g., showing refugees massified, crossing borders, in refugee camps or shelters), opting instead for dignifying, individual portraits set against plain backgrounds. Future research could explore how various contextual cues, such as portraying refugees individually or within mass gatherings at borders or in refugee camps, influence the impact of tearful representations on eliciting compassion towards refugees.

The role of depicting refugees in tears in activating anger at the injustices they face from host societies requires further investigation. Data from the three online experiments that focused on moral anger yielded inconsistent findings, and no significant effects were unveiled in the meta-analytical integration. Echoing some prior evidence (Bobowik et al., 2022; Hendriks et al., 2008), Study 2 participants exhibited moral anger in response to tearful refugee expressions, unlike those in Study 1. This discrepancy may stem from subtle differences between both studies in contextual cues in which the emotion expression was embedded.[23] Compared to Study 1, in which the narrative about the experiences of the refugee protagonist might have been appraised as mostly sad, likely evoking feelings of compassion, Study 2, lacking such a narrative, allowed for the tears to be interpreted as expressions of either sadness or frustration (see Vingerhoets, 2013 for the definitions of "sad crying" and "protest crying"), thus eliciting reactions ranging from compassion to moral

[23] In Study 2, we replaced one of the items from Study 1 ("offended") with a more appropriate item to reflect moral anger ("indignation"). However, when the analyses were conducted without the differing items, results remained the same.

anger or even encompassing both. These results underscore the importance of considering the interplay between the visual content representing refugees and their personal stories (that is, the social context). It opens avenues for further research into identifying narratives that most effectively resonate with host societies, fostering a deeper engagement with the refugees' emotional experiences.

Further, it would be interesting to investigate how various narratives, paired with tearful images of refugees, translate into different forms of help. Previous research indicates that anger induction increases people's willingness to financially contribute to causes addressing equity concerns (e.g., offer compensations to empower women affected by human trafficking, van Doorn et al., 2017). However, the activation of anger did not significantly influence their readiness to participate in charitable acts aimed at easing women's immediate suffering. It is thus plausible that moral anger may play a more pronounced role in encouraging autonomy-promoting aid towards refugees, particularly when participants encounter images and stories that symbolize "tears of protest". For instance, narratives focusing on the denial of refugee status or the challenging living conditions in refugee camps or centers could potentially amplify such reactions.

Concerning self-oriented emotions, our data shed some light on the ongoing debate on the two possible pathways of reactions to others' suffering, including empathic and non-empathic emotions (Batson et al., 1997; Cialdini et al., 1997). Whereas members of the receiving society clearly show other-oriented empathic reactions to emotional images communicating refugees' helplessness and neediness, as a result they might also experience self-oriented sadness (Küster, 2018), but not necessarily feel distressed, perhaps because exposure to visual representations does not involve direct contact with the suffering person. This finding aligns with some previous studies which did not find support for the role of tearful expressions among migrants in eliciting observers' discomfort (Bobowik et al., 2022),

Jointly, helping intentions and support generated by depictions of refugees with visible tears seem mostly altruistic and not egoistic in nature (see Batson et al., 1987).

The Impact of Tears on Intergroup Helping Intentions and Behavior

Existing research suggests that the visual framing of refugees in the media may translate into intentions to offer refugees help (e.g. Borah & Irom, 2021). The main novelty of the current research is the examination of the influence of exposure to tearful faces of refugees on *intergroup* help, including real and meaningful helping behavior (donations), as well as two types of helping intentions, namely, the readiness to offer refugees immediate help to alleviate their suffering (dependency-oriented help) and the willingness to engage in supporting refugee rights and thus ensuring their agency (autonomy-oriented help, e.g., Louis et al., 2019; Nadler & Halabi, 2006).

The noteworthy finding, supported by the meta-analytical integration of the data, is that the exposure to tearful images of refugees was linked with actual behavior (money donated to a cause supporting refugees), *indirectly* via an affective mechanism: the activation of compassionate feelings in response to tearful representation of refugees. Our research extends existing evidence on the effects of tearful expressions on people's real-life prosocial behavior, beyond sharing resources in a fictitious economic game (Reed et al., 2019). That is, our research demonstrates that in response to tearful images of refugees, people give away money (earned for participation in the experiment in Studies 1, 2, and 4 or their own money in Study 3) to support organizations that work to share refugee stories and thus strive to make their voices heard and improve their quality of life, because they feel concerned with the refugee's suffering and support actions aiming at sharing their stories. It should be, however, highlighted that we relied on intimate, humanizing portraits of single individuals, which differs from the way refugees are typically depicted in mass media, that is, in medium to

large groups, without recognizable facial features, and with medium-to-long distance camera shots (Azevedo et al., 2021; Bleiker et al., 2013).

We also corroborated our predictions that depicting refugees as expressing their sadness in tears is associated with people's willingness to assist refugees with their immediate reception by providing dependency-oriented aid, via the compassionate feelings towards the protagonists — as confirmed by the meta-analytical integration of the data. Our research thus offers evidence in support of an empathy-helping link in response to other's suffering (e.g., Batson et al., 1981, 1983; Dovidio et al., 1997), and specifically on the relationship between empathic emotions and dependency-oriented aid towards refugees (Knab & Steffens, 2021).

Further exploration of our data provided evidence that compassionate feelings in response to tearful displays are also associated with higher intention to offer autonomy-oriented help through giving refugees access to education, professional training, or political life, and these effects were comparable in size to those observed for more immediate dependency-oriented help. In addition, the data from the three online experiments showed that members of the host society exposed to tearful refugees donate more money to an organization supporting refugees than those who are shown more neutral images of refugees to the extent they experience compassion in response to these images and therefore are willing to take part in autonomy-oriented prosocial actions in support of refugees. There are several plausible explanations for this. First, our data does not support the idea put forward by some scholars that compassion only serves to motivate costless and superficial forms of help, such as offering some comfort to terminate the distressful situation (Cialdini et al., 1997; Neuberg et al., 1997). In our research, the more the participants experienced compassionate feelings towards the refugee, the more they were willing to engage not only in those "smaller" gestures of support (such as providing emotional comfort or offering clothing

and blankets), but also in other forms of help, such as offering asylum seekers employment in their profession or guarantee their rights of political participation, which might be perceived as much more costly for members of the Dutch society. These results clearly converge with intergroup relations research showing that compassion is linked with equity concerns and collective action oriented at granting disadvantaged groups equal rights and support (e.g., Iyer et al. 2003; Saab et al., 2015). One needs to consider that in our research both types of helping intentions were strongly correlated, and perhaps also for this reason similar affective mechanisms were involved in explaining the effects of tears on both types of helping intentions. Finally, it is also important to note that our data did not provide evidence for the direct impact of tearful displays on any of the measures of intergroup helping, and future research should further test to what extent expression of suffering by vulnerable outgroup members generalizes to intergroup helping beyond the specific situation in which the expressing occurs.

Exploratory analyses additionally showed that, across the four studies, refugees' tearful expressions were related to helping intentions and behavior via a sequential process involving emotion recognition and emotional reaction towards the refugee(s). Specifically, in line with the empathy-altruism hypothesis (Batson et al., 1981, 1987) and prior empirical evidence (Bobowik et al., 2022; Zickfeld et al., 2021), members of the host society reacted to tears with compassion and engaged in helping to the extent that they perceived refugees as suffering (i.e., as experiencing sadness).

Limitations

We acknowledge some limitations that provide suggestions for future research. First, in two out of four studies, we relied on interview material with a single woman asylum seeker from Syria, and thus our results cannot be extrapolated to refugees of any type, gender, age, and origin. To develop a more comprehensive picture of the impact of tearful images of

refugees on solidarity, it is necessary to diversify the targets displayed in the research (see Judd et al., 2012). We addressed this issue by including several portraits of refugees in a field, exhibition-based study, and by randomly exposing our participants to one of several target images of both genders in Study 4.

Another limitation involves the issue of causality of the relations tested in the structural model. Although our predictor in the model was experimentally manipulated, the mediators and the outcome measures relied on correlational data. Thus, we cannot draw firm conclusions about the direction of influence between felt emotions and the outcomes and future studies should try to experimentally manipulate these emotions (Pirlott & MacKinnon, 2016). It is also worth noting that the obtained indirect effects, especially in Study 1, were small, mostly due to the weaker manipulation. Thus, our results regarding the indirect effects need to be interpreted with caution. However, in line with recommendations for appropriate analyses of indirect effects (Fiedler et al., 2018), our statistical tests are grounded in solid a priori theoretical knowledge and reasoning explaining the mediating role of emotion responses in the link between the occurrence of tears and helping. Moreover, we acknowledge that, apart from emotional responses, there are alternative explanatory mechanisms, including cognitive inferences about criers (e.g., Bobowik et al., 2022; Reed et al., 2019; Zickfeld et al., 2021) or their perceived emotional experience, such as sadness or remorse (e.g., Küster, 2018; Picó et al., 2020).

A common drawback of field studies is that it is often not possible to ensure the random assignment of participants; in our case, to the tearful and non-tearful conditions. However, a major strength of our research is testing the link between emotional tears and behavior. In three out of four studies participants were invited to offer a donation to a specific, relatively unfamiliar organization (Refugee Today) and a specific cause (bringing focus on refugees by telling their situation and stories). In Study 4, we could address this

limitation and found similar effects when participants had an opportunity to donate to a better-known organization (i.e., UNHCR) and choose between supporting two different causes, one oriented towards providing more immediate assistance and another towards enhancing refugees' autonomy.

Conclusion and Practical Implications

This research reveals that the exposure to portrayals of refugees in tears is, to the extent that these images activate feelings of compassion, linked with people's willingness to offer refugees assistance as well as donating money to solidarity initiatives. We provided this evidence based on robust, nation-wide samples from online experiments and a field citizen science research project involving a photography exhibition of tearful and non-tearful portraits of refugees. Importantly, our research has created an opportunity for launching fundraising actions that allow us to generate real resources to support initiatives aimed at sharing stories of displaced individuals and thus empower the refugee community. Collectively, our findings raise important questions regarding the nature and broader implications of the visual framing of refugees' emotional experiences for different forms of prosocial behavior, and open new avenues for participatory research on this topic, bridging art, social and emotion science, and activism.

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